

UNIVERSAL MULTIMODAL CO-GENERATION WITH CONTINUOUS BITSTREAM DIFFUSION

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ABSTRACT

[Luca will lead the pitch and framing of the paper; to be written last, once results are final.]

1 INTRODUCTION

❗ **Micha:** See my suggested intro below

Continuous-state generative diffusion models (Ho et al., 2020; Song et al., 2021b) have achieved strong results in image (Rombach et al., 2022b), video (Ho et al., 2022), and audio generation (Kong et al., 2021). These models generate samples by learning to reverse a corruption process that progressively transforms data into noise. Their success has motivated extending diffusion to discrete domains, including language (Sahoo et al., 2024) and protein sequences (Wang et al., 2024b).

For discrete data, widely studied approaches use discrete corruption processes, including masking (Sahoo et al., 2024) and categorical token replacement (Austin et al., 2021). An alternative is to represent discrete tokens in a continuous space and apply Gaussian diffusion, as explored by Diffusion-LM (Li et al., 2022) and continuous diffusion for categorical data (Dieleman et al., 2022). Analog Bits provides a particularly simple representation: encode discrete values as binary strings and treat their bits as continuous variables during diffusion (Chen et al., 2023a). Recent continuous diffusion and flow-based language models report sample quality competitive with, and in some settings better than, discrete diffusion baselines (Chen et al., 2026; Hu et al., 2026). These results motivate investigating continuous diffusion as a shared modeling framework for heterogeneous modalities.

Multimodal generation requires capturing dependencies between observations with different statistical and structural properties, such as images and captions, speech and transcripts, or protein sequences and structures. Existing diffusion and flow-based systems already support joint and conditional generation in image-text settings (Bao et al., 2023b; Tang et al., 2023b) and protein sequence-structure settings (Campbell et al., 2024; Wang et al., 2025b). Building on this progress, we investigate how far a common representation, denoising objective, and data-dependent calibration rule can transfer across these domains, with modality-specific choices concentrated in tokenization and data layout.

In this paper, we extend continuous bitstream diffusion (CoBit) from language modeling to multimodal co-generation (Batzolis et al., 2026a). Frozen modality-specific tokenizers map observations to discrete codes, which are organized into a shared bitstream and corrupted with continuous Gaussian noise. For each modality combination, a single denoiser is trained on joint and conditional tasks through partial denoising: observed positions remain clean while the remaining positions are corrupted and reconstructed. The same weights therefore support joint generation and the conditional directions included in training. The extension introduces a shared multimodal code space and a partial-denoising training recipe while retaining CoBit’s corruption process, denoising objective, and backbone family.

Different modality combinations can lose information at different noise levels, motivating data-dependent calibration. Following text-only CoBit, we estimate a conditional-entropy-rate profile from denoising errors during training and use it to guide training noise allocation (Raya et al., 2026) and the placement of inference steps (Stancevic et al., 2025). Related work has extended this entropy-rate view of inference scheduling to flow-matching and Schrödinger-bridge models (?). The resulting schedule provides a common calibration rule across domains. Tokenizer configura-

tions, sequence layouts, and inference-time settings such as guidance, stochasticity, and step count remain task-dependent.

We evaluate this framework on image–text, speech–text, and protein sequence–structure tasks. On zero-shot MS-COCO evaluation, the 462M-parameter image–text model achieves a text-to-image FID-30K of 18.48 with 20 denoiser evaluations, while also supporting captioning and joint generation from the same weights. The 633M-parameter speech–text model achieves a 3.7% word error rate between co-generated text and Whisper transcriptions of the generated speech. Protein experiments assess folding, inverse folding, and generated-backbone designability. Together, these experiments examine the transfer of a shared bitstream diffusion recipe across domains and identify the task-dependent performance and sampling trade-offs that remain.

2 RELATED WORK

(...)

Continuous bitstream diffusion and entropy-based scheduling. Diffusion models for discrete data commonly use making or categorical corruption, while an alternative is to embed discrete variables in continuous space and apply Gaussian Diffusion (Austin et al., 2021; Sahoo et al., 2024; Li et al., 2022; Dieleman et al., 2022). Analog Bits (Chen et al., 2023a) represents discrete values directly as binary string with continuously diffused bits, approach which has been extended to language modelling by CoBit (Batzolis et al., 2026a). Multimodal diffusion and flow models have meanwhile explored joint and conditional generation across image–text and protein sequence–structure modalities (Bao et al., 2023b; Tang et al., 2023b; Campbell et al., 2024; Wang et al., 2024b).

Information-theoretic diffusion scheduling. Entropic time scheduling for diffusion models was first introduced by (Stancevic et al., 2025), allocating sampling steps according to the rate at which information is lost along the diffusion process. (Raya et al., 2026) subsequently used related entropy-rate quantities to allocate noise levels during training. More recently, (Trentini et al., 2026) generalized this information-theoretic perspective beyond diffusion, deriving entropy-rate schedulers for flow and bridge generative models.

3 METHOD

Our model is the continuous bitstream diffusion model of Batzolis et al. (2026b) (CoBit), a text-generation framework, applied *unchanged* to sequences that interleave several modalities. The corruption process, the denoiser parameterisation and its loss, the noise schedule and its estimator, the self-conditioning mechanism, the sampler and the backbone are those of the text-only model, with the same hyperparameters. What is added is small and lives entirely in the data pipeline: (i) a way of writing any set of tokenised modalities into one shared binary vocabulary, so that a single network reads and writes all of them through one interface (§3.2); and (ii) a training recipe under which one set of weights supports unconditional co-generation of all modalities present, conditional generation in every direction between them, and any mixture of the two, with the direction chosen at inference time by deciding which positions are held fixed (§3.3). There are no modality-specific encoders, decoders, heads, embedding tables or losses, and no modality-specific training stages. We recap the base model first, to the extent needed to make these two additions precise.

3.1 CONTINUOUS BITSTREAM DIFFUSION

Bits as the universal representation. CoBit models a sequence of N discrete tokens $t_1, \dots, t_N$ from a vocabulary of size V by writing each token as its $m = \lceil \log_2 V \rceil$ -bit binary index, giving a flat bitstream $\mathbf{x}_0 \in \{0, 1\}^S$ with $S = Nm$. Nothing else about the data is used: the framework does not know whether a token is a word piece, an image patch code, a speech frame or a residue, and it never sees a token identity, only its bits. Once written as bits, the sequence is treated as continuous data. A standard variance-exploding Gaussian diffusion is run on it, $\mathbf{x}_\sigma = \mathbf{x}_0 + \sigma \epsilon$ with $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\sigma \in [\sigma_{\min}, \sigma_{\max}] = [0.002, 80]$, taking the data centre to be $1/2$ and $\sigma_{\text{data}} = 1/2$. A denoiser $D_\theta(\mathbf{x}_\sigma, \sigma) \in (0, 1)^S$ predicts the posterior probability that each bit is one and is trained

with the EDM-weighted denoising loss (Karras et al., 2022) in probability space,

$$\mathcal{L}(\theta) = \mathbb{E} \left[w(\sigma) \frac{1}{S} \|D_\theta(\mathbf{x}_\sigma, \sigma) - \mathbf{x}_0\|^2 \right], \quad w(\sigma) = \frac{\sigma^2 + \sigma_{\text{data}}^2}{\sigma^2 \sigma_{\text{data}}^2}, \quad (1)$$

which induces the score $\nabla \log p_\sigma(\mathbf{x}_\sigma) \approx (D_\theta - \mathbf{x}_\sigma)/\sigma^2$ used by the sampler. The network is not asked to learn the local Gaussian denoising problem: the analytic posterior logit of an isolated bit under a uniform prior, $(\mathbf{x}_\sigma - \frac{1}{2})/\sigma^2$ (clipped to ± 30), is added to a learned contextual residual, so that the network models only the dependence *between* bits—which is where all modality-specific structure resides. Three components of the base recipe matter for what follows.

Entropy-derived noise schedule. The rate at which information about $\mathbf{x}_0$ is destroyed as σ grows depends on the data, and different modalities—and mixtures of modalities—lose their information at different noise levels. Rather than fixing a proposal over σ by hand, CoBit measures this rate on the data being modelled and allocates both training noise levels and sampling steps according to it. The conditional entropy rate per unit $\log \sigma$ is estimated online from the unweighted per-example denoising error, $\hat{h}(\sigma) \propto e(\sigma)/\sigma^2$ with $e(\sigma) = \frac{1}{S} \|D_\theta - \mathbf{x}_0\|^2$, kept in a FIFO buffer over recent training examples (8×10^5) and binned into 128 log-spaced bins. The training proposal for σ , and through its inverse CDF the sampling grid, is $q(\sigma) \propto g(\sigma; c, n) \hat{h}(\sigma)^{1/2}$, where $g(\sigma; c, n) = \sigma^n / (\sigma^n + c^n)$ ($c = 0.1$, $n = 3$) is a smooth gate removing mass from very small σ . Training starts from the EDM log-normal proposal ($P_{\text{mean}} = -1.2$, $P_{\text{std}} = 1.2$), ramps linearly to q over steps $4\text{--}5 \times 10^4$, and refreshes q every 2×10^3 steps thereafter. Because $\hat{h}$ is measured rather than assumed, the schedule is self-calibrating: the same estimator with the same hyperparameters is used for every modality combination, and it adapts itself to each.

Self-conditioning. Following Chen et al. (2023b), the denoiser takes a second input $\hat{\mathbf{x}}_0$, an estimate of the clean bits, concatenated bitwise with $\mathbf{x}_\sigma$. During training, with probability $p_{\text{sc}} = 0.5$ a first no-gradient pass produces $\hat{\mathbf{x}}_0 = D_\theta(\mathbf{x}_\sigma, \sigma, \mathbf{0})$, which is detached and fed to the second, trained pass; otherwise $\hat{\mathbf{x}}_0 = \mathbf{0}$. At sampling time two refresh modes are available. In `carry` mode the probabilities returned at step i are passed as $\hat{\mathbf{x}}_0$ to step $i+1$, so self-conditioning costs no additional network evaluations; in `refined` mode an extra denoiser pass is run at the new state $(\mathbf{x}_{\sigma_{i+1}}, \sigma_{i+1})$ conditioned on the carried estimate, doubling the evaluations per step. We use `carry` throughout unless stated otherwise.

Sampler and backbone. We integrate the probability-flow ODE $d\mathbf{x}/d\sigma = (\mathbf{x} - D_\theta(\mathbf{x}, \sigma))/\sigma$ with a first-order (DDIM/Euler) step (Song et al., 2021a) on the entropy-derived grid, optionally with EDM-style churn (re-noising by a factor $1 + \gamma_i$ before each step; Karras et al., 2022) as the stochastic variant. Integration stops at a terminal $\sigma_{\text{dec}} > \sigma_{\text{min}}$; the final probabilities are thresholded at $1/2$, regrouped into m -bit codes and handed to the appropriate tokenizer’s decoder. The number of steps, the guidance scale (§3.3) and the churn are inference-time choices that require no retraining. The denoiser is the Sequence Diffusion Transformer of Batzolis et al. (2026b): the m bits of a position, with their per-bit positional features, form one patch that is linearly projected to one trunk token, so the transformer runs at the token length N rather than the bit length Nm ; the trunk uses RoPE, SwiGLU and adaLN-zero conditioning on σ , and a light skip-connection head combines the trunk output with the raw noisy bits to produce the m per-bit logits of each position. The same backbone is used for every modality combination; only its depth and width vary between checkpoints.

3.2 MULTIMODAL BITSTREAMS

The base model needs nothing from its input except that it be a sequence of fixed-width binary codes. This is what makes it modality-agnostic, and it makes extending it to new modalities a data-preparation step rather than a modelling one. Three ingredients are required.

Modality-specific tokenizers. Each modality is mapped to a run of integer codes by a discrete tokenizer of its own—a byte-pair vocabulary for text, a vector- or lookup-free quantiser for images or audio, a residue alphabet or a structure quantiser for proteins—and back by the matching decoder. The tokenizers are the *only* modality-specific components in the system. They are frozen, they are not trained or fine-tuned jointly with the diffusion model, and they sit outside it: the diffusion model never sees an embedding, a spectrogram or a pixel, only bits. Any tokenizer that yields fixed-width codes can be used; quantisers whose codebook is the full set of sign patterns of a latent vector (lookup-free or finite-scalar quantisation; Yu et al., 2024; Mentzer et al., 2024) pair especially

naturally with bitstream diffusion, since every bit string is then a valid code and the bits are the quantiser’s own latent coordinates rather than an arbitrary index. The tokenizers used for each modality pair are stated in the corresponding experimental subsection (§4).

One shared code space. The per-modality vocabularies are placed in disjoint ranges of a single integer code space, together with a small reserved block of control codes (padding and segment markers), and m is chosen as the bit width of that union—in practice by offsetting the vocabularies or, equivalently, by prepending one or more namespace bits that select the modality. The m bits of a position therefore determine both the modality and the content of a token, every position is a “token of any modality”, and the patch size of the backbone is m for all of them. Consequently there are no modality-specific embedding tables, encoders, heads or losses, and the shared Bernoulli output head costs $O(m) = O(\log V)$ per position regardless of how many vocabularies are merged. The one modality-aware parameter we add is a learned segment embedding: a zero-initialised vector per segment type (control, text, image, audio, ...) added to the per-bit trunk input, which tells the network which segment a bit belongs to without introducing any per-modality computation.

A fixed sequence template. A training example is laid out as a fixed-length template of segments, one per modality, each of a fixed token budget (truncated or padded), separated by structural markers (start/end of sequence and of each segment). Markers are ordinary codes in the reserved block; they are always presented clean, at training and at sampling, and are never denoised. Two-dimensional or higher-dimensional token grids are flattened along a locality-preserving (Hilbert) curve so that 1-D adjacency reflects spatial adjacency. Modalities that are aligned position by position (for instance a residue identity and a structure token describing the same site) may also share one position by concatenating their codes into a single m -bit word; this too is only a layout choice. *[Confirm against the final protein codec in §4.5: the current stub describes an 18-bit position formed by a 5-bit residue code and a 13-bit DPLM-2 structure code.]* The transformer sees only the flat bitstream; the same code path decodes each segment through its own tokenizer’s decoder.

What is, and is not, modality-specific. Across every experiment in this paper the corruption process, the σ range and σ_{data} , the loss and its weighting, the entropy-schedule estimator and all of its hyperparameters, self-conditioning, the sampler family and the backbone architecture are identical. *[Verify against the final audio and protein training configurations before submission; this sentence is the paper’s central claim and should be checked field by field.]* What changes between modality pairs is confined to the data pipeline: the tokenizers and their decoders, the segment layout (which modalities, in which order, with what token budgets), and the list of training regimes of §3.3. The only quantities chosen *per direction* at inference are the sampler settings (step count, guidance scale, churn), which are free hyperparameters of any diffusion sampler and involve no retraining and no architectural difference.

3.3 GENERATIVE DIRECTIONS

A sequence built as above is partitioned into segments $S_1, \dots, S_K$ (one per modality, plus markers). Any generation task over these modalities is a choice of a *conditioning set* $\mathcal{C}$ of segments that are given and a *target set* $\mathcal{T}$ that must be produced: $\mathcal{C} = \emptyset$ is unconditional joint co-generation of everything; conditioning on a text segment gives text-to-image or text-to-speech; conditioning on an image or a speech segment gives captioning or transcription; conditioning on a prefix of a segment gives continuation; and so on. CoBit implements every such task with the same network and the same weights through a single mechanism, which we call *partial denoising*: the positions in $\mathcal{C}$ are held at their clean values x_0 throughout, and diffusion acts only on the positions in $\mathcal{T}$. Because the model is a bidirectional denoiser over the whole sequence, a clean segment is simply part of the input it conditions on; no task token, task-specific head or encoder is involved.

Training. For each training example we draw a task from a small list of *regimes*—conditioning sets with sampling probabilities—and construct the noisy input accordingly. Let $M \in \{0, 1\}^S$ be the bit mask of the conditioned positions (the segments in $\mathcal{C}$, always including the markers). The input to the denoiser is $x_\sigma = M \odot x_0 + (1 - M) \odot (x_0 + \sigma\epsilon)$, i.e. the conditioning segments are clamped clean rather than noised, and the loss equation 1 is averaged only over the denoised positions $1 - M$. When self-conditioning is active the clean conditioning bits are also written into $\hat{x}_0$ on the masked positions, so the network never has to predict what it is given. To enable classifier-free guidance (Ho & Salimans, 2021), with probability $p_{\text{uncond}} = 0.1$ the conditioning segments (but never the

Modality	Tokenizer	Quantization Scheme	Vocabulary Size
<i>Class-conditional ImageNet</i>			
<i>Image and Text</i>			
Image	OpenMAG-VIT2 (Luo et al., 2024)	LFQ	—
Text	o200k	Byte-level BPE	200,018
<i>Audio and Text</i>			
Audio	StableCodec (Parker et al., 2025)	FSQ	46,656
Speaker	BiCodec (Wang et al., 2025a)	FSQ	4,096
Text	o200k	Byte-level BPE	200,018
<i>Protein Sequence and Structure</i>			
Amino-acid	—	—	32
—	DPLM-2 (Wang et al., 2024a)	LFQ	—

Table 1: Overview of tokenization methods used across the different experiments.

markers) are replaced by the null value $1/2$ —the data centre and the point of zero information for a bit under Gaussian corruption—so that the network also learns the unconditional denoiser for each target set. The regime list is the whole specification of which directions a model supports: adding a direction for a given modality set means adding an entry to this list, without touching the model. Each experimental subsection states the regimes used. Examples whose conditioning was dropped are excluded from the entropy-rate buffer of §3.1, so the schedule reflects the conditional denoising problems the model is actually asked to solve.

Sampling. At inference the task fixes M and the clean values on the conditioned positions; the sampler of §3.1 initialises the target positions from noise, clamps the conditioned positions to their clean bits before every denoiser call, zeroes the ODE velocity on those positions after every call (so they never move) and clamps them once more in the returned probabilities. Guidance uses the standard two-branch estimate in probability space: a conditional branch sees the clean conditioning, an unconditional branch sees the null value $1/2$ on the same positions (markers protected), the two are evaluated as one batch of size $2B$, and the guided estimate $D_g = D_u + w(D_c - D_u)$ replaces D_θ in the score, where $w = 1$ recovers the plain conditional model and $w > 1$ sharpens it. In `carry` mode each branch carries its own self-conditioning estimate. Joint co-generation ($\mathcal{C} = \emptyset$) uses a single branch, so its cost per step is one network evaluation while every guided conditional direction costs two; we report the number of function evaluations (NFE) accordingly. Since the direction is decided entirely by M , the remaining sampler choices—step count, guidance scale, churn—can be set per direction on the same weights, and nothing in the framework requires them to agree across directions. The same weights therefore act as an unconditional co-generator, as a family of conditional generators in every direction seen in training, and as a model that fills in an arbitrary subset of positions given the rest.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

[Shared setup common to all modalities.]

4.2 IMAGE GENERATION (CLASS-CONDITIONAL IMAGENET)

[Georgios to write: image-only results and comparisons.]

4.3 IMAGE AND TEXT

Images and captions are where unified generative models are most often measured, and also where the ingredients that usually carry them—a text-pretrained backbone, a frozen text encoder, curated or re-captioned data—are hardest to separate from the modelling choice itself. We therefore ask what the bitstream formulation transfers with none of them present. On an alt-text corpus, though, every available metric is a proxy: FID and CMMD compare distributions, CLIP compares embeddings,

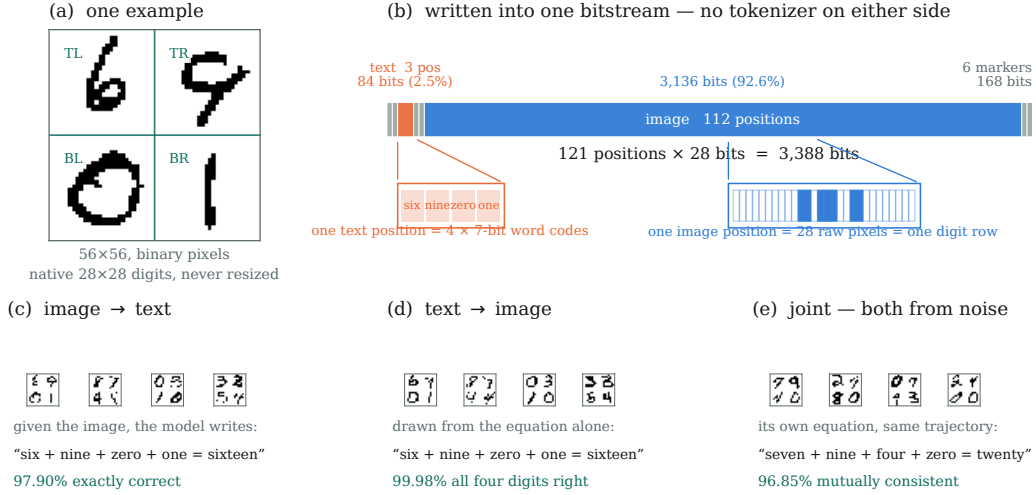


Figure 1: MNIST-Sum. (a) Four native 28×28 binarised digits on a 56×56 canvas, never resized. (b) Image and text in a single 3,388-bit stream of $121 \text{ positions} \times 28 \text{ bits}$, with no tokenizer on either side—one text position holds four 7-bit word codes, one image position holds 28 raw pixels, exactly one row of one digit. (c–e) The three directions, with samples and their exact-match rates at $n=4,096$.

CIDEr compares n -grams, and none of them can say whether a co-generated *pair* actually agrees. We begin instead with a small corpus on which every direction has a unique correct answer.

MNIST-Sum: a ground truth for every direction, and no tokenizer on either side. The corpus pairs an image of four binarised MNIST digits (LeCun et al., 1998) in a 2×2 grid with the raster-order equation and its sum, so each of image $\rightarrow$ text, text $\rightarrow$ image and joint co-generation has a unique correct answer (Figure 1). Neither modality is tokenized: binary pixels ride as raw bits and the text uses a hand-specified 7-bit vocabulary, so the framework is exercised with no modality-specific component at all. A 63.3M-parameter model trained on the CC3M recipe unchanged writes the correct equation for a given image 97.90% of the time against a 6.52% modal-sum baseline, and places all four digits correctly for a given equation in 99.98% of samples. The direction that matters is joint co-generation, where nothing is given and both halves come from one trajectory: **96.85%** of co-generated pairs carry an equation exactly correct for the image produced beside it. Held-out digit tuples score indistinguishably from training ones (Appendix A.7).

Alt-text setup. We train CoBit-CC3M (462M, 24×1024) on Conceptual Captions 3M (Sharma et al., 2018) ($\approx 2.9\text{M}$ pairs) and CoBit-CC12M (633M, 26×1152) on Conceptual 12M (Changpinyo et al., 2021) ($\approx 11.7\text{M}$ pairs, trained to 87% of its planned step budget). Images are tokenized by a frozen Open-MAGVIT2 lookup-free quantiser (Luo et al., 2024; Yu et al., 2024) into 18-bit codes, 16×16 ($f16$) for CC3M and 32×32 ($f8$) for CC12M; text uses o200k, and a 19th namespace bit separates the two, so every position of either modality is one 19-bit token (§3.2). Neither model sees MS-COCO (Lin et al., 2014): every number below is zero-shot under one fixed protocol (Appendix A.1). Text-only pre-training, tokenizer surgery, auxiliary alignment losses and staged curricula are deliberately absent; each is known to help, and we omit them so that the result measures what the unmodified framework transfers.

Results and placement. Table 2 reports all four directions for both models. Text $\rightarrow$ image is the strongest, at zero-shot FID-30K **18.48** (NFE 20) and **16.59**; neither is near the floor of that column, where two disjoint sets of real COCO images already score 2.35 and the codec adds only 0.59 and 0.20, so the distance to larger systems lies in the model and not in the token space—Figure 5 shows the same thing by eye. The text-emitting directions are weaker, at CIDEr 0.136 and image $\rightarrow$ text CLIP 24.83. Joint co-generation is harder again (FID 50.19): with no prompt the model samples its own CC3M-like distribution rather than COCO’s, and the distributional metrics charge it for the difference—but pair coherence 25.67 is about as high as a conditional caption’s agreement with

Direction	n	Steps (NFE)	FID↓	CMMD↓	CLIP↑	CIDEr↑
<i>CoBit-CC3M — one set of weights (462M, f_{16}, 256 image tokens)</i>						
text → image	30,000	10 (20)	18.48	18.41	27.73	—
image → text*	5,000	256 (512)	—	—	24.83	0.136
joint, 10 / 256 steps	30,000	10 / 256	50.19 / 61.95	41.53 / 46.80	23.10 / 25.67 [†]	—
<i>CoBit-CC12M — one set of weights (633M, f_8, 1,024 image tokens)</i>						
text → image	30,000	16 (32)	16.59	17.76	26.51	—
image → text*	5,000	256 (512)	—	—	22.98	0.057
joint, 20 / 256 steps	5,000	20 / 256	56.18 / 64.47	37.60 / 40.33	23.19 / 24.01 [†]	—
<i>Floor of the FID column: two disjoint real draws</i>						
	30,000		2.35	—	—	—
<i>the same draw after the codec, f_{16} / f_8</i>						
	30,000		2.94 / 2.55	—	—	—

Table 2: Zero-shot MS-COCO, all four generative directions from a single set of weights per model. Each row is quoted at the step count best for that direction (*Sampler* below); joint generation has no single optimum, so both ends of its trade-off are shown. CLIP is ViT-B/32. The last two rows are the floor of the FID column: a disjoint draw of real COCO images against the same reference, and that draw after the tokenizer round-trip, so the codec’s own share is 0.59 / 0.20 (Appendix A.1). *CLIP and CIDEr come from separate runs at their own guidance (3.0 / 2.0) against the COCO reference set and the Karpathy test split respectively. [†]Pair coherence between the generated image and its caption, not comparable to the conditional CLIP scores; FID of the joint rows is of the generated image half. FID is biased in n (+5.3 at $n=5,000$), so the CC12M joint rows are not FID-comparable to CC3M’s. Guidance, churn, ViT-L/14, SWD and precision/recall are in Appendix A.2.

Model	Params	Pre-train pairs	FID-30K↓	GenEval↑
<i>CoBit (ours; from scratch, no pretrained components)</i>				
CoBit-CC3M	462M	2.9M	18.48	0.249
CoBit-CC12M	633M	11.7M	16.59	0.222
<i>Same pre-training corpus (CC3M), zero-shot COCO</i>				
LAFITE (Zhou et al., 2022)	75M (+CLIP)	3.3M	26.94	—
<i>Unified image+text models</i>				
CoDI (Tang et al., 2023a)	—	400M	22.26	0.31
SEED-X (Ge et al., 2024)	17B	—	14.99	0.49
LWM (Liu et al., 2024)	7B	—	12.68	0.47
UniDiffuser (Bao et al., 2023a)	952M	LAION-5B (subsets)	9.71 [†]	—
Show-o (Xie et al., 2024)	1.3B	35M	9.24	0.68
<i>Text-to-image specialists</i>				
DALL·E (Ramesh et al., 2021)	12B	250M	27.50	—
LDM (Rombach et al., 2022a)	1.4B	400M	12.63	0.37
SD v1.5 (Rombach et al., 2022a)	0.9B	2000M	9.62	0.43
PixArt- α (Chen et al., 2024)	0.6B	25M	7.32	0.48

Table 3: Zero-shot MS-COCO FID-30K and GenEval against published systems, as printed in the cited papers and the compilation tables of Xie et al. (2024), Bao et al. (2023a), Zhou et al. (2022) and Ghosh et al. (2023). The rows are not strictly comparable and are not meant to be ranked: parameter counts follow each source’s own convention and exclude frozen components inconsistently, the pair counts count different things, and the same nominal FID moves by about a point between compilations. Appendix A.8 gives the per-row caveats and the systems left out; Appendix A.5 gives the GenEval breakdown and further rows. [†]Bao et al. (2023a) draw 10K and 30K prompts for FID and CLIP score without saying which is which.

its own image (24.83), so the shortfall is in matching the evaluation corpus, not in keeping the modalities together. Table 3 places this among published zero-shot MS-COCO numbers. Against LAFITE, the one system trained on the same corpus under the same protocol, FID improves by 8.46 without its frozen CLIP encoder; among unified image+text models we sit between CoDI and SEED-X, and ahead of the 12B-parameter DALL·E specialist. Every system with a lower FID uses pretrained language or vision components and most train on far more paired data. Compositional

Direction	n	Steps (NFE)	WER↓	CER↓	UTMOS↑	GenPPL-Speech↓
<i>CoBit-LibriTTS — one set of weights (134M, 800 speech tokens)</i>						
text → speech	2,294	512 (1024)	11.7	8.1	3.31	—
speech → text	2,294	512 (1024)	23.3 / 38	15.3 / 24.1	—	—
joint	1,024	256 (256)	19.9	13.3	3.02	377.2
<i>CoBit-MLS — one set of weights (633M, 500 speech tokens)</i>						
text → speech	2,294	512 (1024)	5.3	4.1	4	—
speech → text	2,294	512 (1024)	8.2 / 22	4.3 / 13.1	—	—
joint	1,024	512 (512)	3.8	2.4	3.79	122.5

Table 4: Joint and cross-conditional generative directions from a single set of weights per model for text-audio experiments. Each conditional row is quoted at the best step count for that direction, evaluated on LibriSpeech-PC. Generated audio is transcribed with Whisper-large. Per-direction guidance and churn are listed in Appendix B.1; co-generation uses $\gamma = 0.175$. Text-to-speech evaluation is on test-clean; the two scores for speech-to-text refer to test-clean and test-other, respectively.

prompt following is where the gap is widest (GenEval 0.249 and 0.222, either side of minDALL-E’s 0.23 and far below the LLM-backed unified models; Appendix A.5), and for captioning there is no published zero-shot COCO baseline at all: COCO-trained specialists reach CIDEr 1.02–1.40 against our 0.136. The claim is capable transfer, not a leaderboard entry.

Sampler: the two modalities want step counts $25\times$ apart. Sweeping the step count from 4 to 512 on the same weights, text→image FID and CMMD minimise at 10 steps for CC3M and 16–20 for CC12M and degrade monotonically beyond—the conventional 256 steps costs 4.06 FID for $25.6\times$ the compute—while captioning rises from CIDEr 0.013 at 10 steps to 0.136 at 256; the joint direction shows both effects inside single samples, which is why Table 2 quotes it at both ends. The image-side optimum is driven by the distributional metrics specifically: CLIP moves by under 0.1 between 8 and 256 steps and GenEval by under 0.01 across the same $25\times$ range, so the semantic metrics neither corroborate it nor contradict it. None of this modifies the framework: the step count, like churn and guidance, is an inference-time hyperparameter set per direction on the same weights, and such sampler settings are the only thing that had to be adapted per modality (Appendix A.3).

Scaling, and limitations. Moving to CC12M— $4\times$ the pairs, $\approx 8.7\times$ the compute and a better codec—improves images and degrades text: FID falls $18.48 \rightarrow 16.59$, a margin that survives subtracting each codec’s own contribution (17.90 versus 16.39), while captioning CIDEr falls $0.136 \rightarrow 0.057$ at matched settings. That split is what noisier CC12M alt-text predicts and what uniform undertraining does not, though it remains an inference across metrics rather than a measurement of caption quality (Appendix A.4). The two weaknesses have the same shape. On GenEval, single objects render 80% of the time and colours 46% while position and attribute binding sit near zero, and the score is flat between each model’s image optimum and 256 steps, so it is the model’s ceiling rather than a sampler artefact. Captioning is weak in absolute terms, and the outputs are fluent CC3M-style alt-text scored against COCO’s terse references, which the n -gram metrics penalise almost regardless of content (RefCLIPScore (Hessel et al., 2021) 0.661; Appendix A.6). Both are consistent with CC3M alt-text rarely specifying relations or attribute bindings, and both are exactly what the omitted ingredients are expected to buy. That is the point of the experiment: it shows what unmodified transfer does and does not provide.

4.4 AUDIO AND TEXT

Setup. We train two models: CoBit-LibriTTS (134M, 12×768) on LibriTTS (Zen et al., 2019) (354K pairs, 140K steps) and CoBit-MLS (633M, 26×1152) on both LibriTTS and MLS-English (Pratap et al., 2020) (11.2M pairs, 1.2×10^6 steps, 61% of its planned budget). Both datasets consist of single-speaker reading-style paired samples from the LibriVox audiobook corpus. We quantize audio with the FSQ-based StableCodec (Parker et al., 2025) tokenizer using the 1×46656 configuration that produces 25 codes per second (speech budgets of 800 and 500 tokens), prepended by a fixed-length (32 tokens) global speaker representation from BiCodec-global (Wang et al., 2025a). Speaker tokens are not used during waveform decoding; they are included purely as a conditioning signal for generation. Text is tokenized in the same way as in 4.3 (text budget of 168 and 100

Model	Params	UTMOS $\uparrow$	FSD $\downarrow$	SpkSim $\uparrow$	GenPPL-Speech $\downarrow$	GPT-4o $\uparrow$	SALMON $\uparrow$
<i>CoBit (ours; from scratch, no pretrained components)</i>							
CoBit-LibriTTS	134M	2.82	2,373 / 40.5	0.21	152.1	1.11	78.5
CoBit-MLS	633M	3.89	2,386 / 13	0.34	83.6	1.44	84.3
<i>Trained on MLS-English (same corpus as CoBit-MLS, excluding LibriTTS)</i>							
Flow-SLM-S (Chou et al., 2025)	395M	2.47	3,391 / 35.7	0.21	242.8	1.1	82.83
Flow-SLM-L	1.25B	3.48	3,476 / 35.4	0.41	131.7	1.57	88.25
<i>Speech-only SLMs</i>							
Flow-SLM-Ext.	1.25B	3.47	3,445.1 / 33	0.42	122.4	1.67	90
Llama-Mimi (Sugiura et al., 2025)	1.3B	3.18	3,621 / 14.1	0.35	98.5	1.58	87.92
<i>Text+Speech SLMs</i>							
TASLM-Embed (Tseng et al., 2025)	1.3B	4.16	1,410 / 9.9	0.49	38.6	3.42	78
SpiritLM-Base (Nguyen et al., 2025)	7B	3.2	7,325 / 26.8	0.02	101.4	2	50.5
SpiritLM-Expressive	7B	3.17	6,293.6 / 16.9	0.08	153.2	1.62	58.41
<i>StableCodec</i>	—	4.33	1,119 / 1.26	0.36	—	—	—

Table 5: Continuation-based evaluation for text-audio experiments against published SLMs. All figures are evaluated using publicly available checkpoints, with the exception of generation-based SALMON scores, taken directly from (Hsu et al., 2026). The two FSD records refer to FSD-wlm and FSD-e2v, respectively. More information about FSD and SALMON embeddings is provided on Appendix B.1 We use NFE=512 for both CoBit models. The last row reports the tokenizer performance ceiling.

tokens); the full text-speaker-speech vocabulary can be captured with 18-bit tokens. Along with unconditional, text-conditional and speech-conditional regimes, training also includes speech continuation, where generation is conditioned on the start (3 seconds) of the audio sample; equal effort (0.25) is allocated to each setting. Evaluation is conducted on LibriSpeech-PC test splits and the SALMON benchmark, neither of which have speaker or example overlaps with the training sets. Text pre-training (Hassid et al., 2023) and speaker conditioning on the audio decoder (Wang et al., 2025a; Tseng et al., 2025) are omitted to demonstrate the generality of the unified bitstream diffusion framework. Both modifications are straightforward in principle: warm-initializing CoBit from a pre-trained text checkpoint mainly required adapting the components that are tied to the sequence length and the vocabulary bit-width; speaker conditioning for the decoder is simply a matter of fine-tuning. We expect both changes to close the remaining performance gap and leave their implementation to a dedicated line of work on text-audio CoBit.

Results. Joint generation and cross-conditional directions are reported in Table 4, using the same weights for all tasks. CoBit-MLS can co-generate text and audio consistently, achieving a cross-modal WER of 3.7. Character-level disagreement is even lower at 2.3, and a lot of the additional word conflicts can be attributed to the transcription process: the LM-normalized Whisper-large transcriptions of generated audio can have better syntactic structure than the text produced by the diffusion model, which has considerably smaller language modeling capacity. An UTMOS score of 3.78 suggest natural-sounding audio, even when sampling the speaker identity from scratch as well. The content also remains highly plausible in terms of perplexity under an external LM, despite generation without a text or speech prompt. Both CoBit versions are considerably stronger in text-to-speech synthesis than in speech recognition. In speech continuation, shown in Table 5, both models sample acoustically and linguistically grounded audio, yet they struggle to preserve thematic content.

Comparison to prior work. Table 5 places speech continuation and generation-based SALMON (Hsu et al., 2026) performance among autoregressive speech language models; baseline comparisons for co-generation and cross-modal transfer are offered in Tables 18 and 19, respectively. Per-task SALMON scores are provided in Table 20. Further specification about the comparability of these metrics is provided in Appendix B.6. The capabilities of the bitstream diffusion framework are best shown in speech continuation, where CoBit-MLS outperforms all baselines except TASLM-Embed on UTMOS, FSD-wlm, FSD-e2v and GenPPL. Even CoBit-LibriTTS scores better than the smallest autoregressive model (Flow-SLM-S) across these metrics and matches its weak LLM-as-a-Judge

performance, despite having less than half the parameters and only a fraction of the training data. LLM-as-a-Judge is in general CoBit’s weakest metric; its fixed-rate speech tokenization and from-scratch initialization limit its ability to produce thematically relevant continuations. Even warm initialization, however, appears to be insufficient for semantic extrapolation; thematically accurate speech is only achieved by TASLM-Embed, which uses text-aligned speech representations that close the semantic gap between text and speech modelling. In SALMON-based evaluation, CoBit scores better on average than all text-speech SLMs but cannot match speech-only models. While CoBit remains competitive on voice-related tasks with the aid of its speaker features, it is worse than Flow-SLM and Llama-Mimi models on noise-related tasks; this can be attributed to the Mimi tokenizer (Défossez et al., 2024) they use, which captures more fine-grained details of the audio signal. In text-to-speech, CoBit-MLS surpasses SpiritLM on CER (4.1 to 6.7) and approaches Moshi on WER (5.3 to 4.7), despite being 10 times smaller. However, despite the consistency and naturalness of generated audio, speaker characteristics are not reproduced to the same level as by the ASR-TTS hybrid GPA-v1.5 (Cai et al., 2026) when the decoder is not conditioned on the speaker. Speech recognition is where the absence of text initialization is the most apparent, as it contradicts the design of ASR-specific systems, which typically recognize speech at the character or phoneme level using self-supervised representations, and then employ a text LM to decode their output into well-structured sentences (Radford et al., 2023). For SLMs, the textually pre-trained backbone implicitly plays a similar role, leading to more coherent transcriptions, as previously demonstrated (Maiti et al., 2024).

Sampler. *[Decide what settings need to be considered to match image/biological experiments]*

Effect of model and corpus scale. Increasing the model scale and adding MLS-English to the training set improves performance across all tasks and metrics. MLS-English is much larger (45k hours compared to 580), preserves reading-style audio and is part of StableCodec’s training corpus as well, eliminating the risk of distributional shifts during tokenization. CoBit-MLS is better at aligning text and speech sequences when generating both from scratch, yet CoBit-LibriTTS achieves the best error rates when given the text as input. The performance gap is the smallest for SALMON accuracy, as the smaller model can still capture acoustic properties of the speech prompt. Despite its linguistic fluency and frame-level realism, the main limitation of CoBit-LibriTTS is producing natural-sounding audio, observed across all speech-emitting tasks in terms of UTMOS.

Limitations. Across both models, LLM-as-a-Judge is often at the lowest possible score (Figure X) with averages of 1.11 and 1.44 for CoBit-LibriTTS and CoBit-MLS, respectively. It is very difficult to infer relevant continuations to a speech prompt when the training data is semantically limited to storybook passages. Speech-to-text transfer is also weak, but a large portion of the errors are related only to word-level alignment (WER is almost $\times 2$ CER on test-clean for CoBit-MLS). These limitations are strongly tied to the lack of text pre-training. Imprecise speaker replication in terms of SpkSim can be directly associated with the speaker features being used only in the diffusion process and not in the audio decoder. While all three issues could be accounted for by adjustments that are well established in speech language modeling literature, this work demonstrates what the generalized bitstream diffusion framework can achieve off-the-shelf.

4.5 PROTEIN SEQUENCE AND STRUCTURE

Unlike text, image, and audio multi-modality, whose correspondence is primarily semantic, a protein sequence and its three-dimensional structure are physically coupled descriptions. We ask whether the bitstream formulation can learn the coupled modalities of protein sequence and structure by training CoBit from scratch on 216,237 PDB/AFDB-SwissProt (Berman et al., 2000; Varadi et al., 2024) pairs with 20% sequence replay. Each residue aligns five amino-acids bits with thirteen signs from the frozen DPLM-2 lookup-free quantiser (LFQ) (Wang et al., 2024b) **Micha: quantiser → tokenizer?** **Bruno: that’s the method’s name – LFQ.** Decoding retains the 20 canonical amino acids and all 2^{13} structural codes. The common binary interface lets one denoiser learn sequence-structure dependencies while the frozen codec recovers backbone coordinates. Data, splits, and tokenisation are detailed in Appendices **Bruno: ref appendix.**

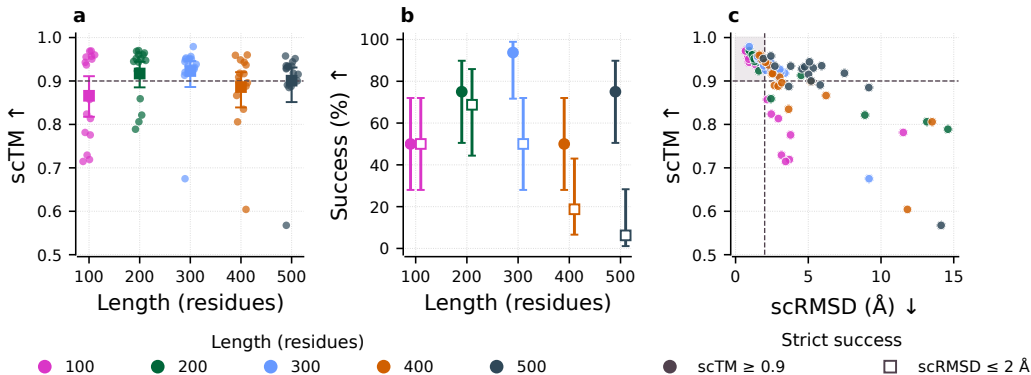


Figure 2: Backbone quality from denoised bits: (a) scTM by length, (b) strict success by length, and (c) fold similarity versus coordinate error. Colours denote length. 80 backbones each receive one ProteinMPNN redesign and ESMFold refold. Intervals are 95%. Shading marks both strict criteria.

Tasks coverage. Clamping either channel gives inverse or forwarding folding while leaving both free engenders joint generation, without task-specific fine-tuning or Multiple Sequence Alignment as input. **Bruno:** cite MSA?. Our evaluations were based on entropic DDIM with 50 grid points with conditional CFG 5 and no unconditional guidance. Under these settings, mean recovery is 44.10% on 128 CATH 4.3 validation targets and forward TM is .554 on 183 CAMEO Targets (Sillitoe et al., 2021; Haas et al., 2018; Zhang & Skolnick, 2004). These results test bidirectional learning through one shared bitstream objective. For 80 generated backbones (16 per length, 100–500), one ProteinMPNN redesign and ESMFold refold give mean scTM .899 (Dauparas et al., 2022; Lin et al., 2023). Figure 2 shows fold similarity across lengths while Figure 3 shows the best backbones per length-bucket. We assess backbone recovery after redesign and refolding using scTM ≥ .9 (D_{TM}) for overall fold agreement and scRMSD ≤ 2 Å (D_{RMSD}) for close coordinate agreement. Full metrics and sampling analysis are in Appendices. **Bruno:** ?.

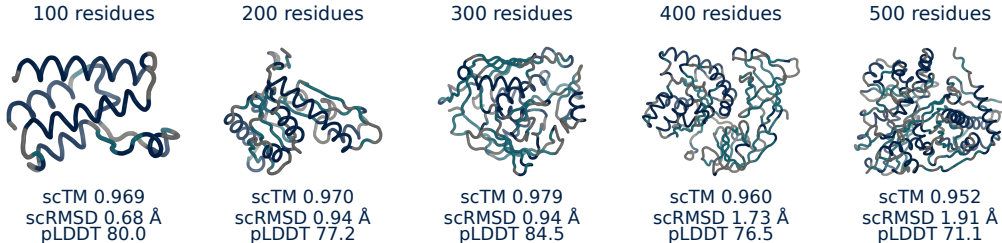


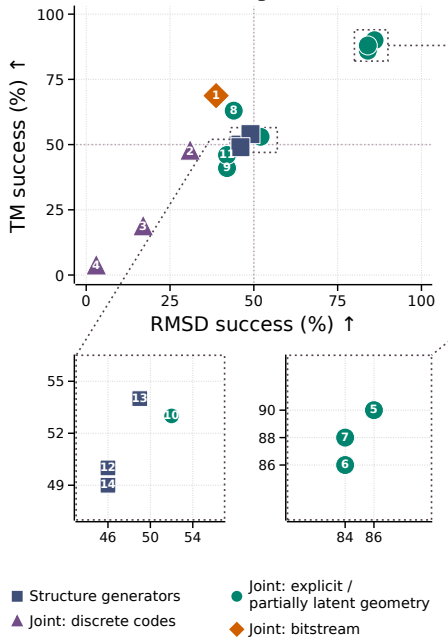
Figure 3: Selected CoBit backbones, rendered with PyMOL (Schrödinger, LLC, 2026). Highest scTM at each length among samples with scRMSD ≤ 2 Å; scores use ProteinMPNN/ESMFold. pLDDT is predicted confidence.

Bitstream capability versus protein optimisation. We compare CoBit’s bitstream formulation against protein-specific models. Table 6 inserts CoBit into the extensive research done by (Lu et al., 2026). MultiFlow (Campbell et al., 2024) leads on the strict criteria with a 21.8M-parameters model due to its explicit geometric learning and redesign-select and refolding-filtered training. On the other hand, CoBit tests the common bit objective with an existing codec and paired-data recipe. Its sequence replay supplies unpaired example unlike MultiFlow’s designability distillation and without direct geometric supervision. **Bruno:** Appendix-X explains the training differences and MultiFlow’s distillation ablation in detail.

Table 6: Backbone designability (%) with one Protein-MPNN redesign and ESMFold. Published baselines: Lu et al. (2026); G: general bitstream framework; P: protein-specific generator.

ID	Model	Scope	n	D_{RMSD}	D_{TM}
<i>Our evaluation: joint bitstream generator</i>					
1	CoBit	G	80	38.8	68.8
<i>Published evaluation: Lu et al. (2026)</i>					
Joint generators: discrete structural codes					
2	DPLM-2 (650M)	P	100	31.0	48.0
3	ESM3 Open (sequence first)	P	100	17.0	19.0
4	ESM3 Open (structure first)	P	100	3.0	4.0
Joint generators: explicit / partially latent geometry					
5	MultiFlow	P	100	86.0	90.0
6	La-Proteina (no-triangle)	P	100	84.0	86.0
7	La-Proteina (triangle)	P	100	84.0	88.0
8	SimpleDesign	P	100	44.0	63.0
9	ProtPardelle	P	100	42.0	41.0
10	ProtPardelle-1c	P	100	52.0	53.0
11	ProteinGenerator	P	100	42.0	46.0
Structure generators					
12	Proteina	P	100	46.0	50.0
13	RFdiffusion	P	100	49.0	54.0
14	FrameFlow	P	100	46.0	49.0

Figure 4: Visual representation of the rates from Table 6. CoBit captures how a proteins folds overall, but is less accurate on the fine details of its shape.



A dedicated protein study can test that coupling and stronger supervision while testing scalability for biological tasks such as binder-linker design with laboratory (wet lab) validation. CoBit’s stronger fold-level (D_{TM}) than coordinate-level (D_{RMSD}) success indicates that denoising structural bits capture useful protein organization information and that geometric precision can be further leveraged in a dedicated protein-training work, but this is highly dependent, in our case, to protein length and training set availability as shown in Figure 2. The present metrics indicate modeling quality, but establishing biological function requires additional setups and experiments.

5 DISCUSSION

[To be written.]

6 CONCLUSION

[To be written.]

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A IMAGE AND TEXT EXPERIMENTS

Supporting material for §4.3. Every figure was read from the job log of the run that produced it. Unless stated otherwise, sampling is deterministic DDIM on the entropy-derived grid with `carry` self-conditioning and $\sigma_{\text{dec}}=0.1$, and sweeps are at $n=5,000$.

A.1 PROTOCOL, CHECKPOINTS AND THE FLOOR OF THE FID COLUMN

Zero-shot MS-COCO. 30,000 captions are sampled with a fixed seed from COCO `val2014` (Lin et al., 2014), one image is generated per caption, and FID (Heusel et al., 2017) is computed with Inception-V3 pool3 features against the 30,000 real images (both resized on the shorter side to 256 and centre-cropped). CLIP score is $100 \cdot \cos(\text{image}, \text{text})$ with OpenAI ViT-B/32 (Radford et al., 2021); ViT-L/14 is also given in Table 8, because published CLIP scores use that backbone and the two scales differ substantially. CMMD (Jayasumana et al., 2024) and SWD are computed in CLIP ViT-B/32 space against a 30,000-image real bank; precision and recall follow Kynkäänniemi et al. (2019) in the same space ($k=3$). Captioning uses the complete Karpathy test split (5,000 images) with `pycocoevalcap` including SPICE (Anderson et al., 2016); CLIPScore and RefCLIPScore follow Hessel et al. (2021). GenEval (Ghosh et al., 2023) uses the official harness (553 prompts $\times$ 4 images). The FID reference statistics and prompt captions are byte-identical between the `f16` and `f8` asset builds, and both models’ image $\rightarrow$ text runs evaluate the same 5,000 images in the same order, so the cross-model comparisons are exact on both sides.

	CoBit-CC3M	CoBit-CC12M
Training corpus	CC3M (≈ 2.9 M pairs)	CC12M (≈ 11.7 M pairs)
Parameters; layers $\times$ width	462M; 24×1024	633M; 26×1152
Image tokenizer (Open-MAGVIT2, 18-bit LFQ)	$f16$: $16 \times 16 = 256$	$f8$: $32 \times 32 = 1,024$
Text budget L_t ; positions; bits S	48; 310; 5,890	96; 1,126; 21,394
Training steps; relative compute	1,000,000; $1 \times$	1,744,000 (87%); $\approx 8.7 \times$
Tokenizer rFID, paired / PSNR (COCO-30K)	1.592 / 22.24 dB	0.726 / 26.92 dB
Codec contribution to an unpaired FID	0.588	0.199

Table 7: The two checkpoints. CC12M has $4 \times$ the data, $1.37 \times$ the parameters, $3.63 \times$ the positions per step and $1.74 \times$ the steps, and a materially better codec, so every image-side comparison in §A.4 is biased in CC12M’s favour. The paired rFID is the Fréchet distance between the 30,000 real images and their own round-trip; the row below it is the codec’s contribution when the two sets are disjoint, which is the quantity comparable to a generative FID.

Operating points. Classifier-free guidance evaluates the conditional and null branches as one batch (§3.3), so $\text{NFE} = 2 \times \text{steps}$ for guided directions and $= \text{steps}$ for joint generation. Text $\rightarrow$ image: deterministic, guidance 2.5 (CC3M) / 2.0 (CC12M), 10 / 16 steps. Image $\rightarrow$ text CLIP: churn $\gamma=0.175$, guidance 3.0, 256 steps. Captioning: churn $\gamma=0.175$, guidance 2.0, 256 steps. Joint: deterministic, unguided. Each was chosen by a bracketed sweep read on CMMD (CIDEr for captioning) and confirmed at the full protocol where one exists. Two repeats of the 256-step headline five days apart gave FID 22.7843 and 22.7881 (CMMD 22.1704 / 22.1627); they differ in world size, so they bound seed and sharding variation together and the ≈ 0.05 FID threshold they set is a conservative one.

Sample-count bias, and the floor. FID is strongly biased in n : at a fixed configuration CC3M scores 28.10 at $n=5,000$ against 22.79 at $n=30,000$ (256 steps), and 24.00 against 18.73 at 10 steps, while CMMD moves by under one point. Sweeps are therefore read on CMMD, FID is quoted at $n=30,000$ wherever a full-protocol run exists, and FID is never compared across different n . FID at this n is also bounded away from zero. A disjoint 30,000-image real draw from COCO train2014 (seed 42, zero image-id overlap; neither model trains on COCO) scores **2.35** against the same reference, 99% of it in the covariance term—finite-sample estimation, not a train/val shift, which would show in the mean. The same draw after a tokenizer round-trip gives 2.94 ($f16$) and 2.55 ($f8$), so the codec adds 0.59 and 0.20, and those are the numbers a generative FID is read against here. The paired reconstruction FID of Table 7 (1.59 / 0.73) compares images with their own round-trip and so omits the finite-sample term that every disjoint-set FID carries, the generative one included; it is a codec statistic, not a quantity to subtract from a generative FID. Neither codec is handicapped by its ImageNet training domain: on a matched 500-image comparison both reconstruct CC3M and CC12M *better* than ImageNet itself ($f8$ LPIPS 0.064 / 0.083 against 0.103, with non-overlapping 95% intervals).

A.2 COMPLETE RESULTS

A.3 SAMPLER CHARACTERISATION

Guidance. CMMD minimises near guidance 2 for both models (CC3M 25.80 / 23.10 / **22.44** / 22.57 / 22.71 / 23.27 at guidance 1.0 / 1.5 / 2.0 / 3.0 / 4.0 / 5.0, 256 steps; CC12M 22.32 / 21.39 / **21.25** / 22.32 / 23.24 at 1.0–4.0), FID one to two notches higher, and CLIP rises monotonically throughout, so past the peak image quality must be judged on CMMD rather than CLIP. Both optima are bracketed on both sides. Guidance and step count are close to separable: re-sweeping at the re-tuned step counts leaves the structure unchanged (CC3M at 10 steps, guidance 2.0 / 2.5 / 3.0: CMMD 18.77 / **18.50** / 18.76; CC12M at 20 steps, guidance 1.5 / 2.0 / 2.5 / 3.0: 18.70 / **17.79** / 17.92 / 18.33), which is why the one-dimensional step sweep at fixed guidance was a valid search. CC12M loses CLIP to CC3M at every matched guidance value (by 0.52–1.02) while winning FID and CMMD at guidance 1–3.

Churn, schedule, self-conditioning. Measured on CC3M at 256 steps, hence far from the image optimum. Churn is strongly harmful for text $\rightarrow$ image—at guidance 3.0, FID 28.10 / CMMD 22.57 deterministic against 36.89 / 34.73 at $\gamma=0.1$ and 41.87 / 40.51 at $\gamma=0.175$, while CLIP moves by at

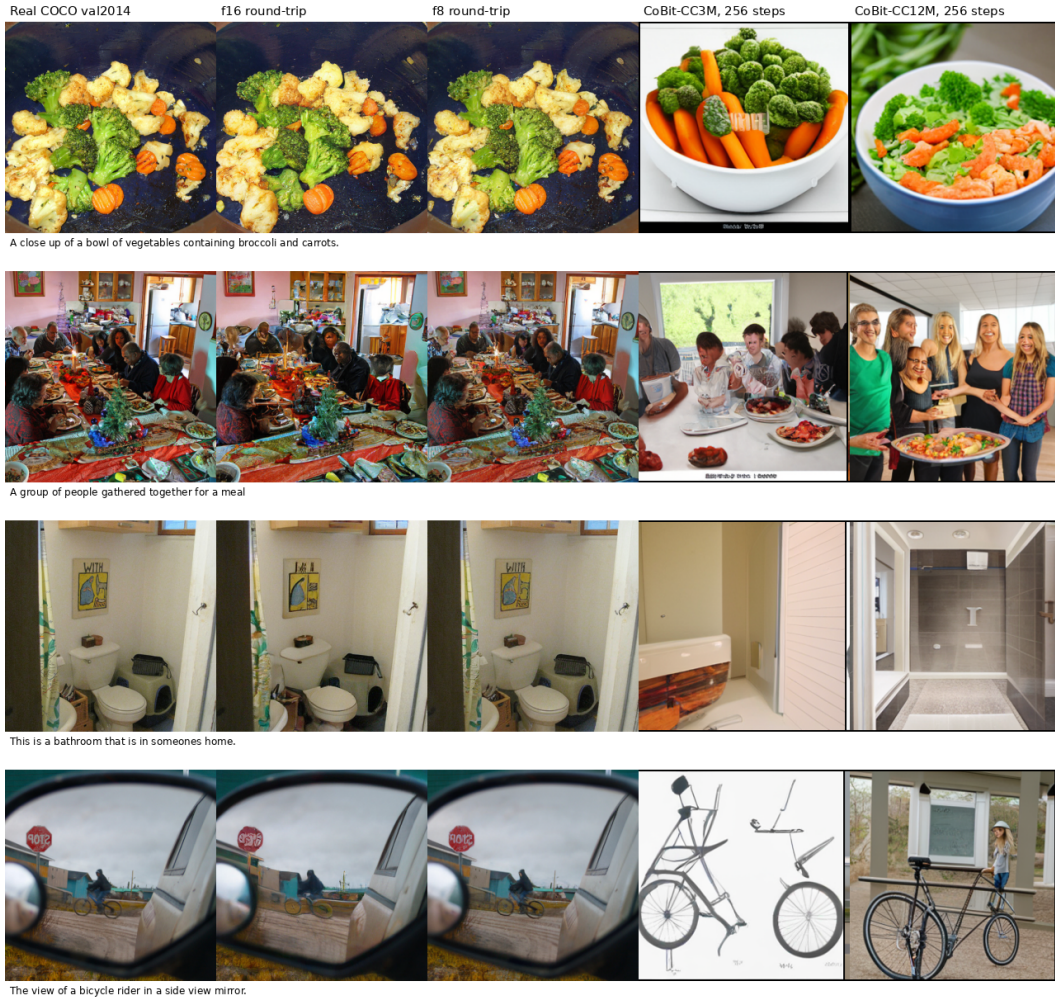


Figure 5: Where the gap is. Left to right: a real MS-COCO val2014 image, its round-trip through the frozen f_{16} and f_8 Open-MAGVIT2 codecs, and text $\rightarrow$ image samples from CoBit-CC3M and CoBit-CC12M for that image’s own caption, at 256 steps. The two round-trip columns are hard to tell from the real image, consistent with a measured codec contribution of 0.59 / 0.20 FID over a 2.35 sampling floor; the visible distance is between the round-trips and the generations. Four of the seven fixed protocol captions, one uncured sample each, illustrating the measured floors of §A.1 rather than standing in for them.

most 0.06—yet the same $\gamma=0.175$ is what makes captioning work at all (deterministic decoding is degenerate, CIDEr 0.018). *Schedule*: for text $\rightarrow$ image the entropy-derived and Karras grids agree at the guidance values actually used, 0.03–0.14 FID apart at guidance 3–5, and separate below them, reaching 0.82 FID in Karras’s favour at guidance 1.0; the two arms ran two days apart on code that logged no sampler configuration, so we do not read that gap either way. For captioning the entropy-derived grid is decisively better, CIDEr 0.136 versus 0.049 at guidance 2.0, so a text $\rightarrow$ image-only ablation would have concluded the schedule does not matter. *Self-conditioning*: `carry` matches refined at guidance 3.0 (CMMD 22.30 versus 22.44, CLIP 27.98 versus 27.99, both at $n=2,000$) at half the NFE, so `carry` is used throughout.

What the semantic metrics cannot see. Four results say the same thing. The CLIP column of Table 9 is flat to within 0.1 between 8 and 256 steps; the CLIP of the guidance paragraph above keeps rising past the point where CMMD and FID have turned; under churn CLIP moves by 0.06 while FID moves by 13.8; and GenEval is step-invariant (§A.5). The embedding and compositional

Direction	n	gs	Steps	FID↓	CMMD↓	SWD↓	CLIP↑		Prec.	Rec.
							B/32	L/14		
<i>CoBit-CC3M</i>										
text → image	30,000	2.5	10	18.48	18.41	0.632	27.73	—	—	—
text → image	30,000	3.0	10	18.73	18.56	0.631	28.03	21.66	0.48	0.28
text → image	30,000	3.0	256	22.79	22.16	0.667	28.02	21.79	0.41	0.29
image → text	10,000	3.0	256	—	—	—	24.86	18.30	—	—
image → text	10,000	4.0	256	—	—	—	24.99	—	—	—
joint (unconditional)	30,000	—	10	50.19	41.53	0.904	23.10	16.87	0.36	0.22
joint (unconditional)	30,000	—	256	61.95	46.80	0.971	25.67	—	0.20	0.30
<i>CoBit-CC12M</i>										
text → image	30,000	2.0	16	16.59	17.76	0.591	26.51	—	—	—
text → image	30,000	2.0	20	16.70	17.68	0.585	26.54	—	—	—
image → text	5,000	4.0	256	—	—	—	23.08	—	—	—
joint (unconditional)	5,000	—	20	56.18	37.60	0.837	23.19	—	—	—
joint (unconditional)	5,000	—	256	64.47	40.33	0.878	24.01	—	—	—

Table 8: Full version of Table 2; the CLIP column of the joint rows is pair coherence. L/14, precision and recall were measured at guidance 3.0 only. Moving CC3M from 256 to 10 steps raises precision 0.41 → 0.48 with recall 0.29 → 0.28, so the gain is not a diversity-for-fidelity trade. Only the L/14 column is comparable with published CLIP scores: UniDiffuser (Bao et al., 2023a) reports roughly 25–26 and 22 on the two directions against our 21.66–21.79 and 18.30, consistent with the FID gap and its LAION-scale pre-training.

Steps	text → image			joint		text-emitting	
	FID↓	CMMD↓	CLIP↑	FID↓	pair-CLIP↑	CIDEr↑	i2t CLIP↑
4	28.55	22.71	27.45	—	—	—	—
6	24.79	19.13	27.86	—	—	—	—
8	24.28	18.81	27.99	—	—	—	—
10	24.00	18.76	28.04	56.06	23.16	0.0131	23.91
12	24.03	18.95	28.05	—	—	—	—
16	24.68	19.40	28.05	—	—	—	—
32	26.13	20.85	28.03	59.14	24.71	0.0588	24.34
64	27.21	21.59	28.00	62.66	25.17	0.0488	24.53
128	27.79	22.18	27.96	—	—	0.1304	—
256	28.10	22.57	28.00	—	—	0.1362	24.83
512	—	—	—	—	—	0.1032	24.90

Table 9: CoBit-CC3M, every direction against the step count, $n=5,000$. The asymmetry is the whole point: text→image minimises at 10 in a bowl bracketed on both sides (8–14) and climbs monotonically above it, captioning peaks at 256 and keeps a tenth of its CIDEr at 10, and the joint direction moves both ways at once because both halves come from one trajectory. The CLIP column is flat to within 0.1 from 8 steps on and cannot see the 4.1-point FID swing beside it; the 64-step CIDEr falling below 32 puts run-to-run noise near 0.01. Text→image is at guidance 3.0, captioning at 2.0 with churn $\gamma=0.175$, image→text at 3.0; guided directions cost NFE = $2 \times \text{steps}$, joint $1 \times$. At the full protocol ($n=30,000$) CC3M moves from FID 22.79 at 256 steps to 18.73 at 10, -4.06 at $1/25.6$ the cost. CC12M behaves the same way at guidance 2.0, optimum 16–20: FID 24.97 / 22.21 / **21.68** / 21.96 / 22.22 / 22.81 / 23.58 / 27.22 at 8 / 12 / 16 / 20 / 24 / 32 / 48 / 256 steps, a larger gain over 256 steps (-5.54) than CC3M’s -4.10 on the same n .

metrics cannot resolve image-quality differences that the distributional ones register clearly. Which family better tracks human preference we do not test here; settling it needs a preference model over hundreds of samples, not inspection of a grid.

	CC3M	CC12M	Δ
<i>Text→image at each model's own optimum, $n=30,000$</i>			
operating point	10 steps, gs 2.5	16 steps, gs 2.0	
FID (raw) ↓	18.48	16.59	−1.89
less the codec's contribution (0.59 / 0.20)	17.90	16.39	−1.50
CMMD ↓ / SWD ↓	18.41 / 0.632	17.76 / 0.591	−0.65 / −0.041
CLIP B/32 ↑	27.73	26.51	−1.22 (<i>worse</i>)
<i>Text-emitting directions, matched 256 steps, $n=5,000$</i>			
captioning CIDEr / SPICE ↑	0.1362 / 0.0541	0.0571 / 0.0347	2.4× / 1.6× <i>worse</i>
i2t CLIP B/32, gs 3.0 / 4.0 ↑	24.83 / 24.97	22.98 / 23.08	−1.85 / −1.89
GenEval overall ↑	0.249	0.222	−0.027

Table 10: Scaling verdict at each model’s own optimum. CC12M wins every image-side *distributional* metric — it loses CLIP — and the margin survives floor correction: the shared 2.35 sampling floor cancels and removing each tokenizer’s codec contribution leaves CC12M 1.50 ahead, so the codec explains about a fifth of the raw gap (Fréchet distances are not additive, so this is a reporting heuristic for the order of the confound; no reconstruction-CMMD floor exists, so that margin keeps an unquantified codec advantage). The gap widens towards the optimum: at a matched guidance 2.0 and $n=5,000$ it is 3.27 FID at 32 steps against 1.39 at 256. On the text side CC3M wins everything by margins far outside the ~ 0.01 CIDEr noise. Undertraining predicts a uniform deficit and noisier CC12M alt-text predicts the split observed; the measured pattern is the second. That is an inference across metrics, not a measurement of caption quality — the decisive CMMD-versus-checkpoint curve has not been run.

Model	Params	Single	Two	Count	Colors	Pos.	Attr.	Overall
Show-o (Xie et al., 2024)	1.3B	0.98	0.80	0.66	0.84	0.31	0.50	0.68
SD3	2B	0.98	0.74	0.63	0.67	0.34	0.36	0.62
SD v1.5	—	0.97	0.38	0.35	0.76	0.04	0.06	0.43
LlamaGen	0.8B	0.71	0.34	0.21	0.58	0.07	0.04	0.32
CoBit-CC3M (ours)	462M	0.80	0.12	0.05	0.46	0.04	0.03	0.249
minDALL-E	—	0.73	0.11	0.12	0.37	0.02	0.01	0.23
CoBit-CC12M (ours)	633M	0.68	0.11	0.06	0.45	0.01	0.01	0.222

Table 11: Per-task GenEval; prior work as printed in Ghosh et al. (2023) (Table 2) and Xie et al. (2024) (Table 3). Our rows use the entropy-derived grid, deterministic sampling, guidance 3.0 and 256 steps. The score is flat in the step count: CC3M scores 0.240 at 10 steps against 0.249 at 256 on one execution path at fixed guidance, so only the step count varies there, and CC12M 0.221 at 16 steps against 0.222 at 256 (guidance 2.0 and 3.0 respectively). GenEval therefore moves by under 0.01 over a range on which FID moves by 4.06. Single objects render 80% of the time and colours 46% while compositional binding stays near the floor, as it does for every sub-billion-parameter model here; CC3M alt-text rarely specifies spatial relations or colour-object bindings. The sampler configuration matters materially: an earlier pass of the same checkpoint that had silently fallen back to a Karras grid, at guidance 2.0, scored 0.209.

A.4 CC3M VERSUS CC12M

A.5 GENEVAL BREAKDOWN

A.6 CAPTIONING

No published zero-shot baseline exists for this number. UniDiffuser (Bao et al., 2023a) evaluates image→text with CLIP score only, and L-Verse’s captioning numbers (Kim et al., 2022) are for a model trained on the MS-COCO Karpathy splits. For scale, on the $\times 100$ convention those papers use, L-Verse reaches CIDEr 102.2, M^2 Transformer (Cornia et al., 2020) 131.2 and OSCAR_L (Li et al., 2020) 140.0, all trained on MS-COCO; our 0.136 is roughly $7.5\times$ below L-Verse, not $100\times$. L-Verse’s own no-truncation variant moves CIDEr from 102.2 to 181.6 while BLEU-4 falls from 39.9 to 27.6 on the same model and images, so caption length and phrasing alone swing these metrics by tens of points. That is the failure here (Figure 6): CIDEr’s IDF-weighted n -grams penalise the

Configuration	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	CLIPScore	RefCLIPScore
deterministic, gs 1.0 (<i>degenerate</i>)	0.004	0.054	0.157	0.018	0.012	—	—
entropic, churn $\gamma=0.175$, gs 2.0	0.0291	0.0975	0.2212	0.1362	0.0541	0.612	0.6606
entropic, churn $\gamma=0.175$, gs 4.0	0.0147	0.0695	0.1804	0.0520	0.0247	0.623	0.6678
Karras, churn $\gamma=0.175$, gs 2.0	0.0131	0.0672	0.1792	0.0485	0.0232	—	0.6578

Table 12: CoBit-CC3M zero-shot captioning on the complete Karpathy test split ($n=5,000$, 256 steps, no empty captions). Guidance 2.0 maximises the n -gram metrics and guidance 4.0 the CLIP-space metrics; the two families disagree, so both are reported. The entropy-derived grid beats Karras by $2.81\times$ on CIDEr while RefCLIPScore moves by 0.003. CC12M’s row is in Table 10, where the comparison is the point.



Figure 6: What CIDEr 0.136 looks like. Uncensored CoBit-CC3M image→text samples at the operating point of Table 12. The outputs are fluent CC3M-register alt-text—the last panel reads “*modern furniture for a dining table add a table feel to any space*”—where COCO’s references are terse declaratives of the form “*A dining table with wooden spoons laid out on it.*” The content is usually right and the register is not, which is the failure the n -gram metrics score and the CIDEr-versus-RefCLIPScore split in Table 12 measures over 5,000 images. Six uncensored examples, shown to make that failure concrete.

register mismatch almost regardless of content. We report CIDEr for completeness and read the result through RefCLIPScore.

A.7 MNIST-SUM

We train a 63.3M-parameter 12×512 model for 500k steps on 10^6 examples with the CC3M recipe unchanged, holding out 10% of the 10^4 digit tuples, and evaluate every direction at $n=4,096$ with 256 steps. Image→text is scored by string equality with no instrument in the loop: 97.90% of equations are exactly right, decomposing into 98.14% perception and 99.66% arithmetic with 0.05% malformed output, against a 6.52% modal-sum baseline. On digit tuples never seen in training the model scores 97.92%, indistinguishable from its in-distribution score (difference -0.02 , 95% CI $[-0.65, +0.60]$), so the mapping is learned rather than memorised. Text→image, read by a quadrant classifier whose ceiling on real composites is 99.78%, places all four digits correctly in 99.98% of samples. Joint co-generation reaches 96.85%, decomposing into 99.98% well-formed equations, 99.78% in which the named addends match the generated image, and 97.02% in which the arithmetic is valid—so the residual error is symbolic rather than cross-modal.

A.8 COMPARABILITY AND MEASUREMENT HYGIENE

Literature figures. Every number in Table 3 was read from the cited paper’s own table or from the compilation tables of Xie et al. (2024), Bao et al. (2023a) and Zhou et al. (2022). Improved VQ-Diffusion (Tang et al., 2022), our closest architectural relative, is excluded because neither of its figures (FID 8.44 on MS-COCO, 15.58 on Conceptual Captions) is a zero-shot COCO number; CogView (Ding et al., 2021) is excluded because its 27.10 blurs MS-COCO, translates the captions into Chinese and re-ranks 60 samples per caption. Three conventions flatter different rows. *Parameters* are as each source counts them and frozen components are included inconsistently: PixArt- α ’s 0.6B excludes a 4.3B T5 encoder and LAFITE’s 75M counts only trainable weights, while other sources fold a frozen encoder into the figure they print. DALL-E’s 27.50 is itself a best-of-512

The task is evaluating the relevance and likelihood of the predicted text continuation, given the text prompt. You should also consider whether the meaning of the text continuation is making sense. The text prompt is: "prompt", and the text continuation is: "content"

You must give an overall rating from 1 to 5. The rating guideline is as below:

- 1: The text continuation is very unlikely and irrelevant to the text prompt.
- 2: The text continuation is unlikely and marginally relevant to the text prompt.
- 3: The text continuation is moderately likely and relevant to the text prompt.
- 4: The text continuation is likely and relevant to the text.
- 5: The text continuation is very likely and highly relevant.

You should take the following steps to provide the score:

First: briefly analyse the sample with the above definition.

Second: MUST follow the output format as: I would rate the score as _

Table 13: LLM-as-a-Judge prompt for thematic relevance evaluation of speech continuation. GPT-4o is used as the judge model.

CLIP-reranked number that its own paper does not print; we reproduce it as the compilations give it. Four further rows—minDALL·E, LlamaGen, DALL·E 2 and Imagen—are left out of Table 3 for space; the first two appear in Table 11, and the others report FID 10.39 (Ramesh et al., 2022) and 7.27 (Saharia et al., 2022). *Pre-train pairs* count different things, and CC3M is 3.3M pairs as released against the 2.9M still downloadable when we built it, which is why the LAFITE and CoBit-CC3M rows differ on a shared corpus. *FID* is not one pipeline: Stable Diffusion is 8.59 in one compilation and 9.62 in another, the same model measured twice, which sets the scale on which small differences should be read. Separately, our generations reproduce CC3M’s distribution—stock-photo watermarks, white backgrounds, single subjects—against COCO’s multi-object scenes, and that distance penalises FID independently of image quality.

Schedule audit. If the entropy tables are absent from a run directory the sampler silently reverts to a Karras grid. Every run was audited for the corresponding warning and every number reported here comes from a run with a count of zero; where the fallback did occur, the run either supplies a “Karras” arm of a schedule comparison, in which the fallback is the treatment, or was superseded by a fallback-free repeat whose numbers are the ones printed. The audit is not optional: the unaudited GenEval pass that scored 0.209 had fallen back. Separately, CC3M’s GenEval 0.249 was produced on the eager execution path, which is the path on which a bf16 defect invalidated an earlier set of CC12M scores; it is unaffected, on two controlled pairs—eager versus compiled at 10 steps (0.240 versus 0.235) and 10 versus 256 steps on the eager path (0.240 versus 0.249)—because CC3M’s 256-token sequence does not trigger it.

B AUDIO AND TEXT EXPERIMENTS

B.1 PROTOCOL AND CHECKPOINTS

LibriSpeech-PC. To assess the linguistic properties of generated speech signals, we transcribe them using an external Whisper-large (Radford et al., 2023) ASR model. For co-generation, error rates are computed between the two generated sequences. The fluency of a text sequence is measured by its perplexity under an external language model (in this case Llama-3.2 1B), computed separately for each modality in co-generation and for speech transcriptions in speech continuation. To further determine the conceptual relevance of produced continuations, GPT-4o is employed in an LLM-as-a-Judge setting, assigning each continuation a score from 1 to 5. The system message is taken directly from Tseng et al. (2025) and presented in Table 13. For TTS, voice replication is captured in terms of speaker similarity (SpkSim), computed as the cosine similarity between the outputs of an ECAPA-TDNN (Desplanques et al., 2020) speaker verification model.

Distributional Metrics. For speech continuation, we are also interested in how closely the generated audio follows the distribution of ground-truth continuations. We thus measure Fréchet Speech Distance (FSD), an adaptation of the FID metric to audio. Population statistics are extracted from self-supervised representations of the true and generated continuations, and their Fréchet distance

Direction	Steps (NFE)	γ	gs
<i>CoBit-LibriTTS — one set of weights (134M, 800 speech tokens)</i>			
text $\rightarrow$ speech	512 (1024)	0.23	6
speech $\rightarrow$ text	512 (1024)	0.37	5
speech continuation	512 (1024)	0.15	5
<i>CoBit-MLS — one set of weights (633M, 500 speech tokens)</i>			
text $\rightarrow$ speech	512 (1024)	0.27	5
speech $\rightarrow$ text	512 (1024)	0.27	5
speech continuation	512 (1024)	0.4	5

Table 14: Sampler operating points for text-audio experiments. Steps, stochasticity and guidance settings were chosen based on validation performance on 500 samples from LibriTTS dev-clean.

is calculated. Following previous work (Chou et al., 2025), two variants are considered: FSD-wlm, which is based on embeddings from the 6th layer of WavLM-large (Chen et al., 2022) to measure frame-level match, and FSD-e2v, which is based on expressiveness representations from Emotion2Vec (Ma et al., 2024). FSD metrics do not evaluate the quality of the generated audio; they capture its population-level resemblance to the original continuations.

Generation-based SALMON for acoustic evaluation. The SALMON benchmark is used for acoustic evaluation. In its original form, SALMON creates replicas of short speech recordings where the start remains identical, but some paralinguistic property (sentiment, speaker identity/gender, background, room) is modified midway. SLMs receive an accuracy score related to how often the positive example is assigned higher likelihood than the negative one. Given the incompatibility of likelihood-based evaluation for diffusion models (Theis et al., 2015), we turn to an alternative generation-based SALMON formulation (Hsu et al., 2026). The examined SLM produces a continuation of the shared start of the two examples, and a judge model calculates embeddings for the positive, negative and generated continuations. The SLM then receives an accuracy score based on how often the generated continuation is closer to the positive example than the negative one, in terms of cosine similarity in the embedding space. TITANET (Koluguri et al., 2022) is used as the judge for sentiment, speaker and gender tasks; HuBERT-large-audio set and Wav2Vec2-large-audio set (La Quatra et al., 2024) are used for background and room tasks, respectively.

Operating points. The sampler configurations (integration steps, stochasticity and guidance scale) for each conditional task have been selected based on modality alignment (WER) on 500 samples from the LibriTTS dev-clean set; the final settings are summarized in Table 14 and full validation scores are reported in Tables 15, 16 and 17.

B.2 GROUND-TRUTH SPEAKER FEATURES AND TRAINING-INFERERENCE MISMATCH

For all three conditional tasks, BiCodec-global speaker tokens are part of the condition. For ASR, speaker tokens are extracted from the full speech signal, which is always provided as input. Inference-time speaker tokens are extracted from the audio prompt for continuation and from a held-out recording for TTS to rule out information leakage. This is standard practice for zero-shot TTS. During training, however, speaker features from the full ground-truth audio samples are used for all tasks. This forces CoBit to learn a tighter relationship between speaker features and speech tokens, simultaneously accounting for the variability in BiCodec-global, which does not guarantee that all recordings of the same speaker produce identical outputs. A training-inference mismatch is therefore introduced to facilitate unified training.

B.3 COMPLETE RESULTS

Tables 18 and 19 contain extensive evaluation metrics and baseline comparisons for co-generation and cross-modal directions, respectively; Table 20 details per-task SALMON accuracy. LLM-as-a-Judge score distributions for CoBit-LibriTTS and CoBit-MLS are visualized in Figure 7

γ	CoBit-LibriTTS			CoBit-MLS		
	WER↓	CER↓	UTMOS↑	WER↓	CER↓	UTMOS↑
<i>Steps=256 (NFE=512); gs=5/gs=6</i>						
0.11	11.4 / 12.3	8.2 / 8.7	3.29 / 3.27	7.2 / 7.4	5.7 / 6.1	3.96 / 3.94
0.13	12.2 / 12.6	8.8 / 8.8	3.3 / 3.27	6.4 / 7.9	5 / 6.4	3.95 / 3.96
0.15	11.4 / 11.9	8.2 / 8.7	3.32 / 3.27	6.8 / 7	5.5 / 5.4	3.97 / 3.95
0.17	12.2 / 12	8.9 / 8.5	3.3 / 3.27	5.5 / 6.9	4.3 / 5.1	3.95 / 3.95
0.19	11.6 / 11.3	8.2 / 8.2	3.3 / 3.27	5.4 / 6.6	4.2 / 5.2	3.96 / 3.94
0.22	11.4 / 12	8 / 8.5	3.27 / 3.25	6.5 / 6.6	5 / 5.3	3.94 / 3.96
0.25	12 / 13	8.2 / 9.8	3.28 / 3.29	5.4 / 6.3	4 / 4.9	3.95 / 3.97
0.28	11.9 / 12.2	8.2 / 8.6	3.27 / 3.26	6.1 / 7.4	4.7 / 6.3	3.95 / 3.95
<i>Steps=512 (NFE=1024); gs=5/gs=6</i>						
0.15	11.3 / 12	8 / 8.5	3.29 / 3.28	6.1 / 7	4.8 / 5.6	3.92 / 3.98
0.19	11.6 / 11.5	8.2 / 8.3	3.31 / 3.26	6.1 / 6.6	4.5 / 5.2	3.94 / 3.95
0.23	11.4 / 10.9	8.2 / 7.7	3.27 / 3.25	5.6 / 6.4	4.2 / 5	3.96 / 3.93
0.27	11 / 11	7.8 / 7.8	3.3 / 3.29	5 / 5.6	3.7 / 4	3.94 / 3.94
0.31	11.3 / 11.3	7.8 / 8.1	3.31 / 3.27	6.2 / 5.6	3.7 / 4	3.95 / 3.96
0.34	15.1 / 11.4	12.8 / 8.3	3.3 / 3.29	5.2 / 5.4	4.1 / 4.2	3.95 / 3.95
0.37	11.4 / 11.4	7.7 / 8	3.28 / 3.27	6.1 / 5.8	4.6 / 4.3	3.94 / 3.93
0.4	13 / 12.4	9.4 / 8.8	3.26 / 3.28	5.9 / 5.2	4.7 / 3.8	3.94 / 3.94

Table 15: Validation results for text-to-speech transfer, computed on 500 samples from LibriTTS dev-clean. Scores with guidance scales 5 and 6 are reported for each γ -steps combination.

γ	CoBit-LibriTTS		CoBit-MLS	
	WER↓	CER↓	WER↓	CER↓
<i>Steps=256 (NFE=512); gs=5/gs=6</i>				
0.11	20.1 / 20.5	13.5 / 13.5	8.2 / 8.2	4.8 / 4.9
0.13	20.6 / 20.9	14 / 13.8	8.1 / 8.1	4.9 / 5
0.15	20.2 / 20.6	13.5 / 13.8	8.3 / 8.1	4.8 / 4.8
0.17	20.5 / 20.2	13.3 / 13.4	8.4 / 8.5	4.9 / 4.9
0.19	20.9 / 20.6	13.9 / 13.5	8.2 / 8	4.8 / 4.7
0.22	20 / 20.5	13.6 / 13.3	8.5 / 8.7	5.1 / 5.1
0.25	19.8 / 20.3	13 / 13.4	8.1 / 8.6	5.1 / 5.2
0.28	20.2 / 20.4	13.8 / 13.7	8.2 / 8.3	5 / 5
<i>Steps=512 (NFE=1024); gs=5/gs=6</i>				
0.15	20.3 / 20.1	13 / 13.2	8.2 / 8.3	4.8 / 4.9
0.19	20 / 20.2	13.2 / 13.1	8.5 / 8.2	4.9 / 4.9
0.23	20.1 / 20	12.9 / 13	8.3 / 8.4	4.8 / 5
0.27	20 / 20	13 / 13.2	7.8 / 8.1	4.6 / 4.9
0.31	20.2 / 20.2	13 / 13.1	8.2 / 8.1	4.7 / 4.9
0.34	19.8 / 20.2	12.8 / 12.9	8.5 / 8.4	4.9 / 5
0.37	19.4 / 20.1	12.7 / 13	8.3 / 8.2	4.8 / 4.8
0.4	20.2 / 20.2	12.9 / 13.1	8.3 / 8.5	4.9 / 4.9

Table 16: Validation results for speech-to-text transfer, computed on 500 samples from LibriTTS dev-clean. Scores with guidance scales 5 and 6 are reported for each γ -steps combination.

B.4 SAMPLER CHARACTERISATION

Text-audio co-generation requires higher step counts.

Effect of Guidance Scale.

Effect of Stochasticity.

γ	CoBit-LibriTTS				CoBit-MLS			
	WER↓	CER↓	UTMOS↑	GenPPL-Speech↓	WER↓	CER↓	UTMOS↑	GenPPL-Speech↓
<i>Steps=256 (NFE=512); gs=5/g=6</i>								
0.11	24.7 / 25.4	16.9 / 17.5	2.48 / 2.4	138.1 / 123.4	5.3 / 5.6	2.9 / 3.2	3.89 / 3.93	77.9 / 79.5
0.13	23.7 / 23.8	16 / 16.2	2.48 / 2.42	122.7 / 121.5	5.6 / 5.2	3.4 / 2.9	3.89 / 3.91	77.1 / 76
0.15	23.8 / 23.5	16.1 / 16.4	2.46 / 2.43	123.7 / 116.4	5.6 / 5.6	3.1 / 3.3	3.91 / 3.91	76 / 75.4
0.17	22.2 / 23.3	15 / 16	2.46 / 2.41	117.9 / 117.6	5.5 / 5.4	3 / 3.1	3.89 / 3.89	78.2 / 74
0.19	22.3 / 23.5	15.5 / 16.1	2.47 / 2.4	117.4 / 121.1	5.4 / 5.5	3.1 / 3	3.89 / 3.9	77.4 / 71.4
0.22	22.6 / 22.8	15.8 / 15.9	2.45 / 2.41	112.7 / 110.9	6.3 / 5.8	3.8 / 3.4	3.89 / 3.94	76.3 / 74.5
0.25	22.6 / 24.2	15.5 / 17	2.45 / 2.42	112.3 / 108.2	5.2 / 5.6	3 / 3.1	3.9 / 3.89	71.5 / 70.3
0.28	22.4 / 24.4	15.5 / 18.7	2.45 / 2.39	106.9 / 107.3	6 / 5	3.7 / 2.8	3.92 / 3.91	73.6 / 70
<i>Steps=512 (NFE=1024); gs=5/g=6</i>								
0.15	19 / 22.1	12.8 / 15.3	2.51 / 2.45	102.8 / 100.7	5.4 / 5.5	3.1 / 3.4	3.93 / 3.91	70.6 / 67.5
0.19	19.9 / 22	13.4 / 15.5	2.51 / 2.45	100.5 / 101.6	5.2 / 5.3	2.9 / 3.1	3.92 / 3.91	67 / 65.9
0.23	21 / 22.6	16.7 / 15.5	2.51 / 2.44	93.5 / 92.7	7.1 / 5.5	3.9 / 3.4	3.92 / 3.9	66.8 / 64.5
0.27	22.5 / 23.2	15.9 / 16.7	2.49 / 2.46	93.7 / 92.7	4.9 / 5.4	2.8 / 3	3.9 / 3.89	66 / 66.6
0.31	20.9 / 23.6	14.7 / 16.7	2.47 / 2.43	87.8 / 93.5	5.3 / 5.7	3 / 3.4	3.91 / 3.89	63.7 / 61.9
0.34	21.2 / 25.1	15.2 / 18.5	2.47 / 2.42	89 / 87.7	5.6 / 5	3.3 / 2.8	3.91 / 3.9	63.4 / 58.8
0.37	23.5 / 25	17 / 18.1	2.46 / 2.42	84 / 86.7	5.3 / 6	3.2 / 3.7	3.9 / 3.9	63.4 / 61.5
0.4	21.7 / 24.6	15.3 / 17.8	2.43 / 2.39	82.8 / 86.2	4.9 / 4.9	2.7 / 2.8	3.9 / 3.91	61.3 / 58.7

Table 17: Validation results for speech continuation, computed on 500 samples from LibriTTS dev-clean. Scores with guidance scales 5 and 6 are reported for each γ -steps combination.

Model	Params	WER↓	CER↓	UTMOS↑	GenPPL-Text↓	GenPPL-Speech↓
<i>CoBit (ours; from scratch, no pretrained components)</i>						
CoBit-LibriTTS	134M	19.9	13.3	3.02	509.6	377.2
CoBit-MLS	633M	3.8	2.4	3.79	247.2	122.5
<i>Autoregressive System (SmolLM+Speaker $\rightarrow$ SparkTTS)</i>						
Cascade	640M	3.2	3.1	4.23	122	1,648

Table 18: Joint generation performance, computed by generating 1,024 samples. Speech perplexity for the cascade system is high, as SmolLM produces multilingual sequences or web/code artifacts that are not transferable to speech. We use the same NFE as in Table 4; 256 for CoBit-LibriTTS and 512 for CoBit-MLS.

B.5 BASELINE SYSTEMS

All known speech language models (SLMs) are autoregressive; diffusion models in speech generation are mostly limited to TTS systems or ASR-TTS hybrids.

Multimodal SLMs and Hybrids. The most comparable text-audio baselines for CoBit are models that learn to generate both text and speech. TASLM (Tseng et al., 2025) generates the two modalities jointly, leveraging continuous text-aligned speech representations from the TASTE encoder and implementing separate prediction heads. The semantic equivalence of text units and speech embeddings bridges the gap between audio and text generation, producing linguistically fluent and conceptually consistent audio. Other approaches focus on systems that can be extended to cross-modal tasks. SpiritLM (Nguyen et al., 2025) is trained on interleaved text-audio sequences, and its expressive variant for additional control over pitch and style. SpiritLM models can perform TTS and ASR via task-specific fine-tuning and few-shot prompting. Moshi (Défossez et al., 2024) is a spoken dialogue system that produces text alongside speech, as an inner monologue that steers generation towards intelligible sentences. Its cross-modal capabilities arise from introducing a time delay on the generated stream and inference-time teacher-forcing on the conditioning stream. GPA (Cai et al., 2026) is a multi-task autoregressive hybrid that treats TTS and ASR as conditional language modelling problems. It is the closest method to CoBit in terms of scale, multi-task joint training and initialization from scratch, but it is limited to cross-conditional tasks and cannot perform speech

Model	Params	Text-to-Speech				Speech-to-Text	
		WER↓	CER↓	UTMOS↑	SpkSim↑	WER↓ Clean	WER↓ Other
<i>CoBit (ours; from scratch, no pretrained components)</i>							
CoBit-LibriTTS	134M	11.7	8.1	3.31	0.19	23.3	38
CoBit-MLS	633M	5.3	4.1	4	0.27	8.2	22
<i>Autoregressive SLMs/Hybrids</i>							
GPA-v1.5 (Cai et al., 2026)	505M	3.1	3.1	4.37	0.4	3.4	6.3
SpiritLM-Base	7B	—	6.7	—	—	6	11
Moshi (Défossez et al., 2024)	7B	4.7	—	—	—	5.7	—
<i>StableCodec</i>	—	—	—	4.3	0.52	—	—

Table 19: Text-to-speech and Speech-to-text performance on LibriSpeech-PC. For text-to-speech, evaluation is conducted on the test-clean partition. For each speaker in the dataset, one sample was held out to extract speaker features. Speech-to-text evaluation is conducted on test-clean and test-other partitions separately. As in Table 4, NFE=512 for both CoBit-LibriTTS and CoBit-MLS.

Model	Sentiment	Speaker	Gender	Bg.- Domain	Bg.- Random	Room
<i>CoBit (ours; from scratch, no pretrained components)</i>						
CoBit-LibriTTS	86	88.5	88.5	66	71	71
CoBit-MLS	90.5	93	93.5	73.5	80	75.5
<i>Trained on MLS-English (same corpus as CoBit-MLS, excluding LibriTTS)</i>						
Flow-SLM-S	85	87	92.5	77	86	69.5
Flow-SLM-L	92.5	99.5	99	77	84.5	77
<i>Speech-only SLMs</i>						
Flow-SLM-Ext.	93	98	100	77	86.5	85.5
Llama-Mimi	93	90	95	81	90	78.5
<i>Text+Speech SLMs</i>						
TASLM-Embed	95	100	100	61.5	54.5	57.5
SpiritLM-Base	47.5	52	50.5	50.5	52.5	50
SpiritLM-Expressive	72	54	56	58.5	55.5	54.5

Table 20: Speech continuation performance across the consistency tasks of the SALMON benchmark for generation-based evaluation. Each model generates a continuation for each prompt, and accuracy scores are computed for each task by comparing the cosine similarity of the generated continuation with the positive and negative examples. Different embedding models are used for each task, specified in Appendix B.1. NFE=512 is used for both CoBit-LibriTTS and CoBit-MLS, with the operating points fine-tuned for speech continuation.

language modelling. However, all of these models require task-specific design and modifications for TTS and ASR, in contrast to CoBit.

Speech-only SLMs. In the context of speech generation, additional comparisons can be drawn to textless SLMs, which mainly rely on the powerful Mimi tokenizer to computationally balance the semantic benefits of text generation with fine-grained acoustic details. Llama-Mimi (Sugiura et al., 2025) autoregressively predicts each Mimi layer, leveraging RVQ’s conditional nature: the generation of each code is conditioned on both the previous tokens and the earlier codebooks of the same token. Flow-SLM (Chou et al., 2025) exploits Mimi’s semantic-acoustic combination for cascaded prediction, breaking the strictly autoregressive paradigm: a causal transformer predicts the semantically distilled first RVQ layer; then, unquantized Mimi embeddings are inferred with flow matching, conditional on look-ahead semantic tokens.

Cascaded Co-generation. None of the multimodal systems are evaluated on unconditional co-generation. TASLM-Embed and Moshi are the only baselines that are, in theory, capable of true parallel generation, but they are not trained for such a task. Matching the autoregressive nature of all other baselines, we instead compare CoBit to a cascaded pipeline: text is generated unconditionally with a text LM and speaker identity is sampled from a small transformer; a TTS system then uses

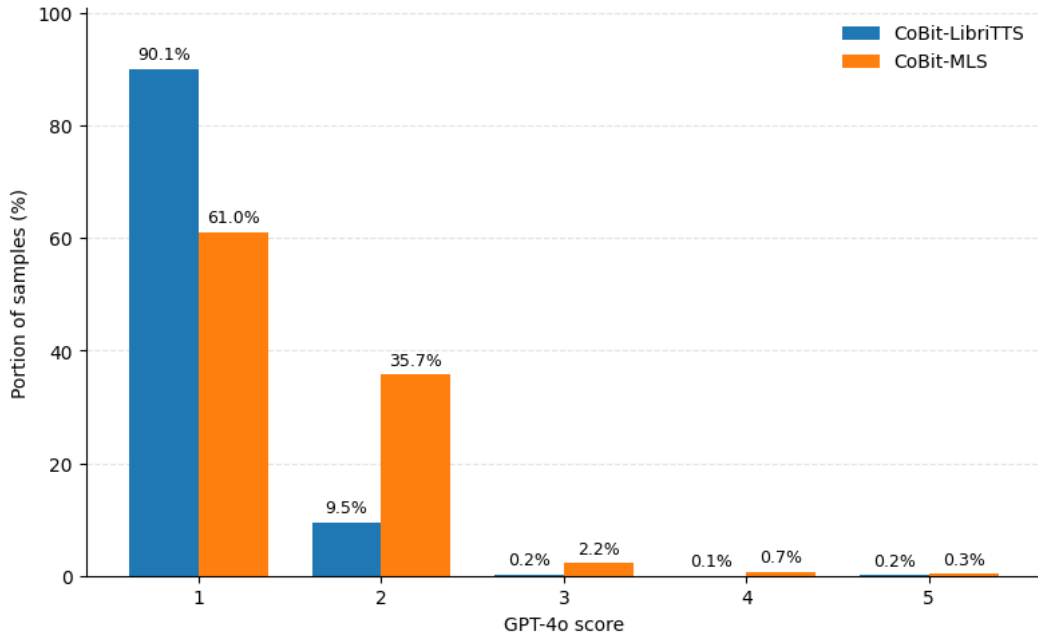


Figure 7: LLM-as-a-Judge score distribution for CoBit-LibriTTS and CoBit-MLS. CoBit-LibriTTS almost always receives the lowest possible score. A large portion of CoBit-MLS samples perform moderately better, but neither of the model can consistently achieve scores of 3 or higher.

Steps	WER↓	CER↓	UTMOS↑	GenPPL-Text↓	GenPPL-Speech↓
16	173.4	120.5	1.51	9,002.5	303.4
24	100	70.4	1.94	5,377.7	440.5
32	68	47.4	2.3	3,665.2	391.4
48	41.7	28.6	2.88	1,857.4	359.5
64	27.8	18.5	3.12	1,267	313.3
128	9.8	6.1	3.61	397.1	207.8
256	4.8	2.6	3.75	250.4	147.2
512	3.8	2.4	3.78	247.2	122.5

Table 21: CoBit-MLS joint generation performance at different step counts, computed by generating 1,024 samples. All rows inject stochasticity at $\gamma = 0.175$. Unconditional generation does not use classifier-free guidance, so each integration step corresponds to a single function evaluation.

the text and speaker features to synthesize the speech sequence. This cascade system effectively targets the same distribution as CoBit for joint generation, factorized as $p(\text{text}, \text{speaker}, \text{speech}) = p(\text{text})p(\text{speaker})p(\text{speech}|\text{text}, \text{speaker})$. To match the 630M parameter scale of CoBit, SmoLLM2-135M (Allal et al., 2025) is used for text generation and Spark-TTS (Wang et al., 2025a) is used for speech synthesis. Speaker features are sampled by a transformer with 1M parameters in the form of BiCodec-global tokens and passed to the Spark-TTS decoder. We train the speaker transformer on LibriTTS and MLS-English. All three components of the cascade are autoregressive. A known drawback is that unconditional generation with a generic text LM can lead to web or code artifacts that cannot be rendered into natural speech, causing very high GenPPL-Speech, as observed in Table 18.

B.6 MEASUREMENT COMPARABILITY

For speech continuation and co-generation, we evaluate all baselines under the same protocol as CoBit using publicly available checkpoints. For cross-modal transfer, we only evaluate GPA-v1.5. While public SpiritLM and Moshi models do exist, their TTS and ASR performance relies on fine-

Steps	Text-to-Speech				Speech-to-Text	
	WER↓	CER↓	UTMOS↑	SpkSim↑	WER↓ Clean	WER↓ Other
16	6.6	4.5	3.85	0.25	8.6	23
24	6.7	4.8	3.96	0.26	8.4	22.8
32	5.7	4	3.99	0.26	8.3	22.7
48	5.5	3.9	4.02	0.27	8.3	22.9
64	6	4.1	4.03	0.27	8.3	22.5
128	6.3	4.1	4.03	0.27	8.1	22.3
256	5.3	5.2	4.03	0.27	8.2	22.4
512	5.3	4.1	4.01	0.26	8.2	22.2

Table 22: CoBit-MLS cross-modal performance performance at different step counts, computed by generating 1,024 samples. All rows inject stochasticity at $\gamma = 0.175$. Classifier-free guidance with $gs=5$ is used throughout the table.

Steps	UTMOS↑	FSD-wlm↓	FSD-e2v↓	SpkSim↑	GenPPL-Speech↓	SALMON↑
16	3.12	2,236.3	8.9	0.3	232	84.41
24	3.53	2,176.6	7.1	0.33	200.9	84.41
32	3.7	2,117.6	6.8	0.34	178.3	84.25
48	3.83	2,127.4	7.5	0.34	150.6	85.08
64	3.88	2,127.4	8.1	0.35	130.2	84.75
128	3.9	2,108.2	9.2	0.34	111.6	85.58
256	3.91	2,108.4	10.6	0.34	94.7	84.41
512	3.91	2,179.3	11.8	0.34	85.8	83.92

Table 23: CoBit-MLS speech continuation performance performance at different step counts, computed by generating 1,024 samples. All rows inject stochasticity at $\gamma = 0.175$. Classifier-free guidance with $gs=5$ is used throughout the table..

tuning and time-delaying tricks (B.5) that cannot be exactly replicated; we therefore report evaluation numbers from their respective papers. All generation-based SALMON scores are taken directly from Hsu et al. (2026). However, the exact protocol documented in the original work is replicated, using the same checkpoints for the judge models.

C PROTEIN SEQUENCE AND STRUCTURE EXPERIMENTS

[Bruno: additional results and experimental details for reproducibility.]

gs	Text-to-Speech				Speech-to-Text	
	WER↓	CER↓	UTMOS↑	SpkSim↑	WER↓ Clean	WER↓ Other
1	8.6	6	3.77	0.25	9.3	23.8
2	5.8	4.9	3.99	0.26	8.2	22.1
3	6.5	5.2	4.01	0.26	8.1	22.2
4	5.4	4.5	4.03	0.27	8.1	22.2
5	5.4	5.2	4.03	0.27	8.2	22.4

Table 24: CoBit-MLS cross-modal performance performance at different guidance scales. All rows use 256 integration steps (256 function evaluations at $gs=1$; 512 for $gs > 1$) inject stochasticity at $\gamma = 0.175$.

gs	UTMOS↑	FSD-wlm↓	FSD-e2v↓	SpkSim↑	GenPPL-Speech↓	SALMON↑
1	3.88	1,953.7	5.1	0.33	172.1	83.41
2	3.94	2,054.1	7.8	0.34	116	85
3	3.92	2,073.1	9.1	0.34	116	85
4	3.92	2,140.3	10	0.34	96.1	84.83
5	3.92	2,108.4	10.6	0.34	94.7	84.41

Table 25: CoBit-MLS speech continuation performance performance at different guidance scales. All rows use 256 integration steps (256 function evaluations at $gs=1$; 512 for $gs > 1$) inject stochasticity at $\gamma = 0.175$.

Direction	WER↓	CER↓	UTMOS↑	SpkSim↑	FSD↓	GenPPL↓	SALMON↑
text → speech	5.9	4.1	3.97	0.26	—	—	—
speech → text	8.6 / 23.4	4.7 / 14	—	—	—	—	—
speech continuation	—	—	3.85	0.33	2,130.4 / 8.2	136.9	85.41
joint	11.3	7.3	3.54	—	—	413.9 / 204.2	—

Table 26: CoBit-MLS scores with a deterministic sampler across all four tasks. Integrations steps are fixed to 256 and the conditional task use classifier-free guidance with strength 5.